TAIKO · PROTOCOL DESIGN · AUGUST 2026

Based Preconfirmation Redesign

A perpetual auction with commit → publish → seal epochs
Design v15 — a learning deck

Anyone can win the sequencing seat · Slashing from day one · Finality at the seal · No oracles

MOTIVATION

Why redesign preconfirmations?

Where we are today

- Preconfirmations run on a **trusted whitelist** of operators — users trust reputation, not collateral.
- The planned permissionless path required preconfers to be **L1 validators** registered in a Universal Registry Contract (URC) — blocked twice over: little validator adoption, heavy integration burden.
- Slashing machinery exists on paper but is **dormant**: bonds are configured to zero, faults have no teeth.
- A preconf promise today is only as good as the operator's brand.

The redesign's premise

- Decouple the seat from L1 validators: sell sequencing rights in an **open, perpetual on-chain auction**.
- Make every promise **bonded and objectively slashable** from the first day.
- Make finalization **carry its own proof** — the seal is one proof-carrying L1 transaction whose acceptance is finality, so there is no separate prover market (G4).
- Design the **failure modes first**: every degraded state has a permissionless exit.

MOTIVATION

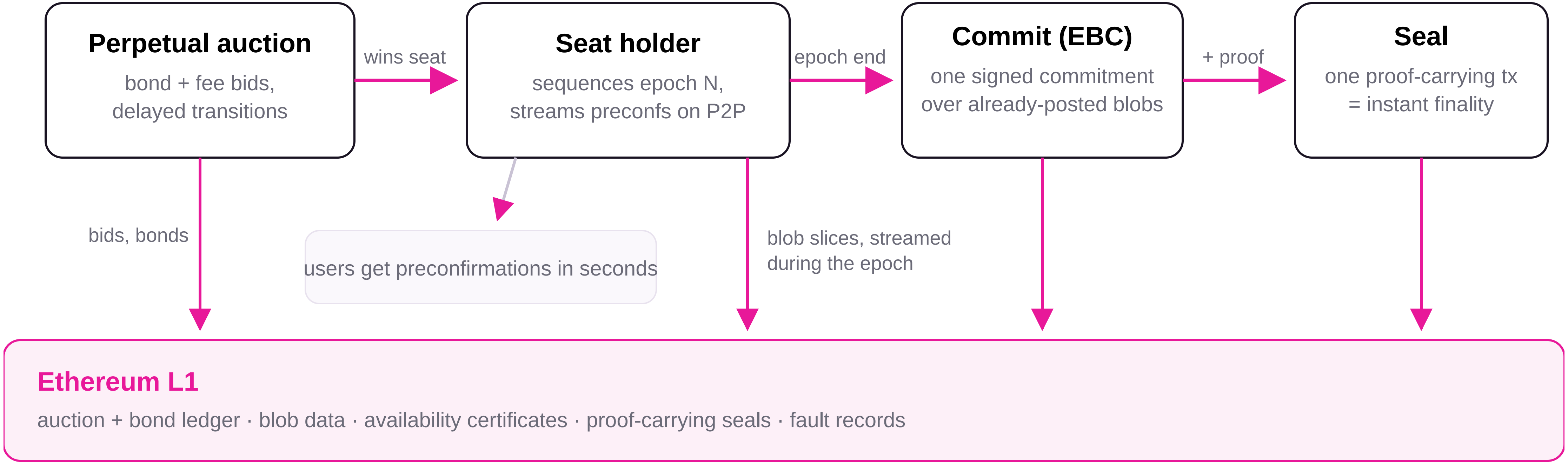
Five goals steer every mechanism

GOAL	WHAT IT DEMANDS
G1	**Anyone can become the preconfer** — a perpetual auction for a single sequencing seat, open to any bonded bidder (no validator requirement).
G2	**Slashing from day one** — on objective faults only: liability is computed from L1 records, never from someone's report.
G3	**Non-validators can act reliably on L1** — every mandatory action has a multi-slot budget; the seal has a multi-epoch window.
G4	**Sealed epochs finalize immediately** — proof verified at acceptance; fast withdrawals; no separate prover market.
G5	**Explicit degradation endgames** — the chain keeps moving through every failure; in the permissionless phase anyone can end a degradation by outbidding the reserve floor.

G5 is a *liveness* guarantee, not a user-fund-safety guarantee: an L1 reorg deeper than the grace κ can revert L2 state exactly as on any rollup.

MENTAL MODEL

The big picture



One seat at a time; one open epoch at a time; every promise backed by bond; every outcome provable by anyone from L1 data — and every no-EBC epoch resolves by a fixed default derivation, so even holderless epochs have one canonical outcome.

CAST

Roles and tenures

- **Seat holder** — the current auction winner; sequences epochs, issues preconfs, commits and seals them.
- **Standby bidders** — bonded runners-up; auto-promoted (bindingly) if the holder quits, idles, or is slashed out.
- **Observers** — anyone; earn bounties for recording faults, submitting availability certificates, and recovering stalled epochs.
- **Users** — receive preconfirmations; can always force-include through L1.
- **Treasury / DAO** — receives auction fees; governs parameters prospectively.
- **L2 nodes** — re-derive everything from L1; hold nobody's word for anything.

Tenure = the accountable identity

A tenure is an immutable id binding the holder, its registered keys, its assigned epochs, and its reserved collateral. Every artifact it signs binds `(chain, tenure, epoch, role)`.

Every fault gets a stable logical id — `H(chain, tenure, epoch, class, position)` — debited **at most once**, ever.

Two-phase rollout

Phase A: auction entry temporarily allowlisted (training wheels; DAO commits to a fast-path SLA). **Phase B:** fully permissionless. Same mechanics in both.

The perpetual auction (on L1)

- One sequencing seat, auctioned **perpetually** — the incumbent keeps it until outbid, quitting, or terminated.
- A bid = **TAIKO liveness bond** + **ETH deposit** + **per-epoch ETH fee** paid to the treasury; a new bid must beat the incumbent by **≥ 10%**. The ETH lives in **one account with a seniority waterfall**: recovery tranche → safety tranche → working tranche, under a single admission solvency check.
- Every transition is delayed by `q = 2` epochs — the current and next epochs are always final, so handovers are never a surprise.
- Exits are never free: quitting gives q epochs of notice; **idling pays exactly what quitting would** (the fee clock keeps running), so going dark is never the cheaper exit.
- Termination on: a settled fault, bond below reservation, or `K_empty = 16` idle epochs — and every tenure **expires after at most T_max** (weeks-scale) and re-auctions: the one idleness/censorship bound no Sybil can reset.

Why the auction lives on L1

The auction and the bond ledger must keep operating even when the L2 itself is degraded or halted — so they live on L1. Slashing *evidence* is proven on L2 (cheap, expressive) and the verdict is bridged up to the single L1 ledger.

No perpetual grandfathering

Every *new* epoch assignment — the incumbent's included — is gated on the *current* reservation. Raise the floor, and a holder that does not top up is gracefully rotated out.

MECHANISM 1 / 4 · COLLATERAL

A split bond — and no oracles, anywhere

Safety bond — in ETH

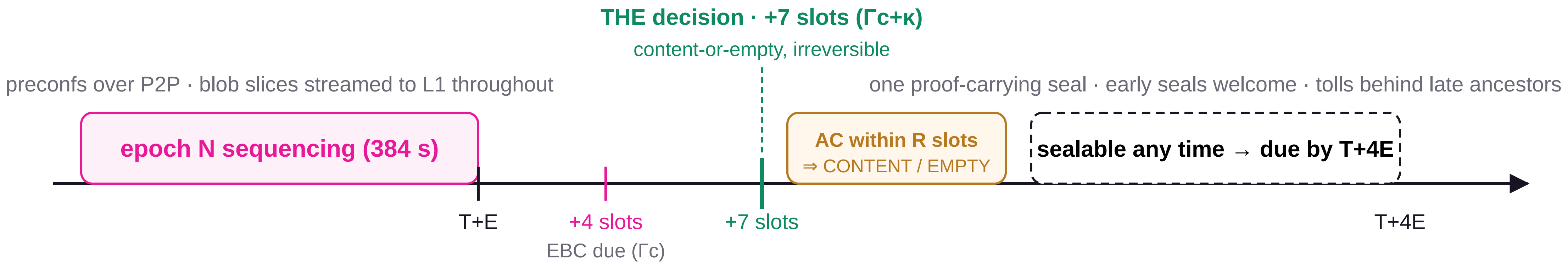
The safety tranche of the tenure's single ETH account. Backs the **theft** class: equivocation / MEV theft, where the attacker's gain is ETH-denominated. Deterrence and gain share the same numéraire, so it is sized to exceed the extractable MEV over the whole adjudication-latency exposure window Λ — **without any price feed** — and its value is fixed from tenure start.

Liveness bond — in TAIKO

Backs the **delay** class: missed commits and seals, which slow the chain but steal nothing. A TAIKO price move weakens griefing deterrence at worst — a bounded, governance-adjustable residual.

- **Owner directives honored together:** TAIKO as the bond token, and no price oracles — ETH is assumed stable-to-increasing.
- A TAIKO price crash cannot cheapen the theft slash; the short-then-misbehave and withdraw-after-shock attacks disappear because the theft debit's value never re-prices.
- Liveness slashes are **≥ 80% burned** — faults must destroy value, not move it.
- All payouts the protocol ever makes go to **precommitted payees** (proof public inputs or pre-registrations) — front-running a copied proof or poke redirects nothing.

One epoch's life on the clock



κ = 3-slot grace: presence at the decision decides — a first inclusion in (D, D+κ] counts; re-posts byte-identical

While epoch N settles, epoch N+1 is already sequencing: its first ~7 slots are **soft** preconfs (parent-provisional), upgrading to **hard** (bond-backed) once N's outcome is *contract-legible* — AC accepted or R-window closed. Proving fits the 4-epoch deadline.

The EBC — one commitment decides the epoch

- The **epoch-boundary commitment** is one-shot per (tenure, epoch) — a malformed or unsigned submission is **inert and never consumes the one-shot** (an explicit acceptance predicate, §3.1). Presence at the single decision (+7 slots) is all that counts: a first valid submission any time in those 7 slots wins.
- It binds the full ordered content, the end-of-preconf tip, and the **KZG commitments of blob slices already posted to L1** — data first, commitment second.
- It also commits **its own L1 origin** (anchor block hash/state root). Derivation reads the committed origin, never the block that happened to include the transaction — so an L1 reorg can shift *when* things land, never *what* they mean.
- The committed anchor must be ≥ 32 slots deep (reorg safety), fresh, and advancing (no serving from stale L1 state).
- Miss the EBC ⇒ the epoch resolves **EMPTY** and a liveness certificate settles. An explicit empty EBC is valid — unless forced inclusions are pending.

Totality: garbage cannot wedge the chain

Derivation is **total and bounded**: any committed bytes map to exactly one L2 block sequence. Malformed or oversized content degrades *deterministically* to default content — under consensus-exact parse caps (size, RLP count/depth, tx count) enforced identically by client and proof circuit.

Consequence: a committed epoch has **one canonical outcome, provable by anyone holding the data**, no matter when or where anything landed on L1.

Publication is part of the commitment — proven on L1

- There is **no separate publication deadline**: the holder streams its blob data to L1 *during its own epoch*, and the EBC is valid **only if every referenced byte is on L1 at the single decision** (+7 slots) — first inclusions and re-posts alike.
- So content-or-empty is a **single decision, 7 slots after the epoch** — not a second judgment ~27 slots later.
- "Nothing new" means nothing outside the EBC's committed commitments — a delayed slice still lands validly after the boundary, from **any** data holder (P2P spreads them during the epoch).
- If a third party fills a slice the holder failed to keep available, the filler earns a reward — **funded by the failing holder**, paid to a pre-registered beneficiary. No fault, no reward, no honest party to front-run.

The availability certificate (AC)

Contracts can't read blobs, so "bytes on L1" is proven in two halves. **On L1:** a permissionless, sender-free AC proves each referenced KZG commitment sits in a canonical beacon block **at or before D+κ** (an SSZ Merkle proof vs an EIP-4788 root), within an R-slot window — first AC ⇒ **CONTENT**, none by R ⇒ **EMPTY**. **In the seal circuit:** the proof opens those commitments and binds exactly those bytes.

Censorship has to beat everyone

The AC is tiny, fee-bumpable calldata anyone with a beacon node can submit — suppressing it means beating every sender for R slots, on top of the slices across a 32-slot span. The most expensive artifact to censor.

MECHANISM 2 / 4 · SEAL

The seal: proving is finalizing

- One small, retryable L1 transaction carries the validity proof for the epoch's canonical outcome; **acceptance = finality** (G4).
- Proofs are fully **precomputable**: content is final 7 slots after the epoch, and the proof binds no later L1 state. The seal has a **deadline only** (due by T+4E, tolled) — sealing early is always valid and shrinks everyone's exposure.
- The seal contributes **zero derivation inputs** — sealing late or from a different sender can never change what the epoch means (I7).
- The payout address is a **public input of the proof**: copying a proof from the mempool advances the chain but pays the original prover's payee (I8).

One open epoch

Epochs seal **strictly in order** through a single `openEpoch` pointer. Every later deadline *tolls* while an ancestor is unresolved — a slow predecessor never converts into a successor's fault.

The recovery lane never closes

At every moment some permissionless action can advance `openEpoch`: the content seal, a forced-only seal, an empty seal, or — as an absolute floor — the mechanical expiry cancellation at `T_exp_eff`. Unproven material never blocks any of them (I3).

THE USER'S VIEW

What a preconfirmation means now

MOMENT	THE PROMISE YOU HOLD	BACKED BY
Seconds — P2P preconf	A signed commitment binding (tenure, epoch, position, commitments, deadline) — equivocation is a safety fault.	ETH safety bond (MEV-sized over Λ)
First ~7 slots of the next epoch	*Soft* preconf: explicitly provisional on the parent's single decision.	Parent's own EBC + bond
Parent outcome contract-legible	AC accepted (CONTENT) or R window closed (EMPTY) — hard, bond-backed preconfs from here.	Protocol invariant (single decision)
At the seal	Proven, finalized L2 state on L1.	Validity proof

Force-included transactions ride the same rails: once snapshotted, an epoch's minimum valid outcome *includes them* — an empty resolution is simply invalid while the forced queue is non-empty (I6). Holderless epochs still get canonical headers from the default derivation rule (§6.8).

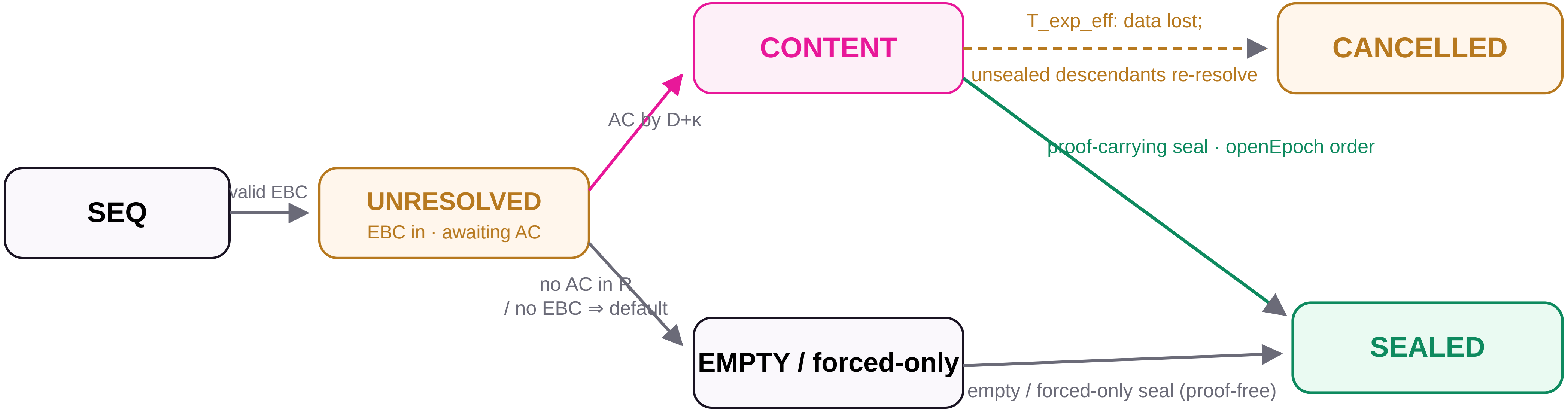
THE CONSTITUTION

Nine invariants everything must obey

- **I1 — Total, bounded, content-addressed derivation.** Committed bytes have one canonical outcome; no-EBC epochs resolve by the default derivation — both independent of all L1 timing.
- **I2 — No fault requires the accused.** Liability is computed state, matured by deadline + absence; every reader materializes it. Pokes accelerate, never gate.
- **I3 — Single open epoch, always advanceable.** A permissionless advancing action always exists.
- **I4 — Successor-safe parent.** Outcomes irreversible at $\Gamma c+\kappa$; deadlines toll behind blocked ancestors.
- **I5 — Bounded global backlog.** Lag > K ⇒ recovery-only mode; the unsealed content tail is structurally capped.
- **I6 — Forced inclusions are censorship-proof.** Non-empty forced queue ⇒ empty outcomes invalid; the forced-only epoch is constructible by anyone.
- **I7 — Only proven transitions lock state; outcomes are sender-free.**
- **I8 — Payees are precommitted, not first-claimers.** Every reward is bound before the witness is public.
- **I9 — Deterrence value is oracle-free.** ETH-matched slashes are ETH-denominated; no mechanism reads a price.
- **Corollary — sealed epochs are final.** No mechanism, including disaster cancellation, ever touches sealed state.

MECHANISM 3 / 4

The epoch state machine



An accepted EBC enters **UNRESOLVED**; the availability certificate (or its R-timeout) picks CONTENT or EMPTY **once**. Holderless / missed / cancelled epochs skip the EBC and resolve by the **default derivation** (§6.8). Seals close epochs **strictly in openEpoch order** — sealed state is untouchable, forever.

Disaster floor: at `T_exp_eff = min(tolled T_exp, blob_slot + retention)` a stuck epoch cancels, marks unsealed descendants VOID (each closes as cancelled, in order), re-queues their forced items, and bills the causing tenure — sealed history is never touched.

MECHANISM 3 / 4 · CENSORSHIP RESISTANCE

Forced inclusion: the floor nobody can lower

- Users enqueue transactions on L1 with a fee `a + b·bytes + c·declared_gas` — L1-measurable quantities with conservative constants that upper-bound proving cost; the per-item base makes flooding pay per item.
- Each epoch takes a deterministic **snapshot** of the queue head; overflow spills to the next epoch.
- While the snapshot is non-empty, the epoch's minimum valid outcome is the **forced-only epoch** — constructible and provable by *anyone* from L1 data alone, with canonical headers from the default derivation (§6.8). Empty is invalid (I6).
- Fees flow to whoever seals the consuming epoch — the fallback funds its own proving.

One lifecycle, no double-spends

Every forced item advances `queued → snapshotted → consumed | refunded` on L1, with its payer stored at enqueue time. Seal proofs take the item's status as a public input; a refund atomically kills every live commitment containing the item. For bridged messages the **msgHash is computed on-chain at enqueue** (`hashMessage`) — never submitter-declared — so a refund can never be steered at a third party's message. Refund *and* execution can never both happen.

Honest accounting

No chain can detect Sybil "demand", so `K_empty` is only a nuisance bound against lazy idlers — the binding bound is `T_max` tenure expiry. Forced-only fee income per tenure is capped, and a holder sealing its own epochs earns no recovery bounty. Squatting has no profit path and a hard clock.

Faults are computed, never reported

- A liveness fault is an **objective L1 fact**: assignment + deadline + absence. It **matures** the instant its deadline-plus-grace passes — no transaction needed, no cooperation from the accused.
- Anyone may record ("poke") a fault for a bounty, but pokes only *accelerate*: every function that depends on fault status — withdrawal, sealing an empty outcome, promotion — **computes maturity itself and materializes the certificate as part of the call**.
- Consequence: censoring pokes buys nothing. A holder that missed an obligation *faults itself by trying to withdraw*.
- Each deadline has exactly **one κ grace** and **one atomic transition**: present ⇒ accepted; absent ⇒ certificate settled, parent outcome fixed, slashability locked — all at the same instant.

Equivocation: two paths to the same slash

A double-EBC (two accepted commits over the holder's own keys) is **slashed directly on L1**. A preconf-vs-record safety fault is proven on L2 by a permissionless watchtower with a **funded accuser reward**; the verdict bridges to the single L1 ledger, which enforces one-debit-per-fault. Verdicts bind a finalized origin root + incarnation — one minted on an orphaned fork dies after a deep reorg.

Withdrawal gate

Exit requires: zero unresolved certificates (computed, not poked), no unsealed epochs, no in-flight verdicts, the tenure's **equivocation-challenge horizon** elapsed (tolled during attested outages), then a ≥ 2-week floor.

When a holder fails, the chain barely notices

The failure playbook

- **Missed commit** → epoch resolves EMPTY at the single decision; certificate settles; slash debits; chain continues.
- **Missed seal** → the epoch's data is on L1, so from the moment the deadline fault matures **anyone** seals it — paid at indexed cost *from the faulter's own recovery tranche*, never from honest tenures.
- **Termination** → the highest standby is auto-promoted (bindingly, after the q-delay); no re-auction gap. No standby ⇒ Total Anarchy (next slides).
- **Every recovery is compensated, never framed**: certificates name only the tenure that actually held the duty; deadlines toll behind blocked ancestors per the normative descendant-tolling rule (§7).

Solvent by admission

The senior tranche of each tenure's single ETH account is sized to its worst-case K outstanding recovery obligations **plus its worst-case cancellation-cascade charge**, checked at admission: `Σ reserves ≥ Σ worst-case obligations`. A deliberate fault can only ever spend its own account; a thin shared pool covers only the honest remainder.

Recovery pays cost, not prizes

Compensation is a capped multiple of prevailing L1 + proving costs (indexed, not fixed) — gas spikes raise it, and there is no margin worth attacking for. Deterrence lives in the ≥ 80% burn, not the bounty.

DEGRADATION

A ladder, not a kill switch

Rung 1 · The seal deadline — fault-paid resolution, nothing global

The moment a stuck epoch's seal deadline passes, its fault matures and anyone seals it from L1 data, paid from the faulter's recovery tranche. One bad holder is never anyone else's problem — and no separate horizon is needed.

Rung 2 · Recovery-only mode — when lag exceeds K = 8 epochs

New discretionary content pauses; all effort funnels into the recovery lane. No withdrawal freeze — honest tenures with clean books may still exit.

Rung 3 · Attested systemic-outage freeze — never reachable by one holder

A global pause requires an independent attestation of a genuine proof-system outage — never openEpoch age alone. It tolls the seal deadline, `T_exp`, *and* the equivocation-challenge horizon on one shared `H_toll_max` clock, so no honest holder is slashed or bankrupted and no equivocator can wait the outage out. Forgiveness never covers whoever drove the stall.

Absolute floor: mechanical expiry cancellation at `T_exp_eff = min(tolled T_exp, blob_slot + retention)` (data genuinely unavailable) — the epoch re-resolves forced-only/empty, unsealed descendants re-resolve deterministically, and the causing tenure pays for the whole cascade. Sealed history is never touched.

Total Anarchy — a fallback that still carries content

- No holder and no standby ⇒ epochs are unowned. The bridge and forced lane keep flowing at L1 cadence — every unowned forced-only epoch gets an advancing, epoch-deterministic anchor from the default derivation (§6.8).
- **Discretionary content is back (owner directive).** Strictly inside a per-epoch **proposal phase** — ending at a mechanical cutoff `T_prop = min(decision + W_a·E, T_F)` — **any address** may submit an **anarchy proposal**: one atomic tx that is *propose* ≡ *seal* ≡ *finality*, with self-contained blob DA and a precommitted payee.
- Two-sided cutoff, the whole repair: proposals valid *before* it, empty/forced-only resolutions *at/after* — closing **both** horns. `W_a = 0` recovers the forced-only/empty fallback exactly.
- **Phase B exit:** anyone ends anarchy by bidding the reserve floor. **Phase A exit:** the DAO's pre-committed fast-path SLA.

Deposits stay safe, and bounded

Even a worst-case voided forced item settles as a refund: queue fees on L1, bridged messages via the terminal-cancellation recall — the source principal returns once proving resumes (an L1→L2-synced `FAILED` mark, `FAILED ⊥ DONE`). Worst-case settlement is a number a depositor can price, not open-ended limbo. Value is only delayed, never burned.

What the lane deliberately omits

No preconfs, no bond, no duty, no equivocation class — a proposal is an *opportunity*, never an obligation. Users get finality-on-landing at proof cadence; their guaranteed path remains the forced queue (I6).

ADVERSARIAL ECONOMICS

Attacks the design prices out

ATTACK	WHY IT FAILS
Preconf equivocation / MEV theft	ETH safety bond sized above extractable MEV over the whole adjudication window Λ; value fixed from tenure start — no oracle, no shorting window, no price-crash discount.
Squat the seat, serve nobody	Idling pays quit-equivalent fees; `K_empty` nuisance-terminates; `T_max` expiry evicts even a perfect Sybil; forced-only income capped.
Withhold the seal, hold users hostage	Data is already on L1; the deadline's matured fault lets anyone seal at the faulter's expense. A stall buys ~minutes, and costs a slash.
Censor the poke / the availability certificate	Liability is computed at read time (I2); the AC is tiny, sender-free, fee-bumpable calldata — suppressing it for R slots buys only a parent EMPTY, never a successor safety-slash.
Cancel-grief a third party's bridge message	The msgHash is computed on-chain at enqueue (`hashMessage`) + a `NEW`-guard on the destination mark — a griefer can only cancel *its own* locked message, never a victim's.
Front-run proofs / pokes / fills / proposals	Payees are precommitted (proof public inputs, pre-registrations) — copying a witness pays the original beneficiary (I8).
Publish a long tail, then cancel it	Recovery-only mode caps the unsealed *discretionary* tail at ≈ K+S epochs (~77 min); forced-only content is data-present and self-funding, not value-at-risk.

Stated residuals: censorship outlasting a holder's whole submission span can cost one liveness slash (priced unprofitable); TAIKO-price erosion weakens only griefing deterrence (the ETH recovery floor is oracle-free per fault); the complete idleness/censorship bound is `T_max` expiry — `K_empty` is only the faster nuisance path.

CONFIDENCE

How hard has this been attacked already?

14 rounds

of adversarial review — ten external rounds (independent AI reviewers, every finding dispositioned publicly) plus a **four-round internal challenge-then-respond loop** (rounds 11–14) and two external bots (Codex, DeepSeek). The loop's severity of *new* findings fell monotonically — **3 → 2 → 0 → 0** IMPORTANT — and a dedicated confirmation pass declared the **loop CLOSED**: no confirmed critical/important defect remains.

1.4 M states

explored at the default bound — **zero invariant violations and deadlock-freedom** (an exit path to finalization always exists), holding **even under a proving outage that never lifts**: the proof-free floor is that exit.

The checker checks itself

24 deliberately-injected design bugs — 17 verification-validity mutants (double decision, ignore-forced, out-of-order seal, ungated withdrawal, missing cascade, double debit, reopen-sealed, …), 4 anarchy-proposal-phase mutants, and 3 fidelity mutants (content-without-AC, late safety-settle, loose tolling) — each caught by the check written for it. **24 / 24 caught.**

BOUND (NEPOCHS × CLOCK)	STATES	VIOLATIONS	HALTS
`2 × 5 (mutation suite)`	1,165,932	0	0 / 0
`2 × 6 (default)`	1,408,548	0	0 / 0

Halts = standard / outage-robust. Exhaustive evidence is `NEPOCHS = 2` today (≥ 3 exceeds memory) — a bounded logical abstraction, not the eventual Solidity.

WHERE IT STANDS

Status and reading map

Before implementation (§13-S)

- The availability certificate (SSZ commitment-inclusion, R window); the default derivation rule and its conformance vectors; the anarchy proposal phase.
- Deep-reorg (> κ) joint-rewind semantics; cascade determinism; consensus-exact parse caps across client and circuit; derivation-input sign-off (Appendix C).
- Recovery-tranche sizing (incl. the cascade charge); forced-item nullifier + L1-authoritative msgHash; verdict incarnation; precommitted-payee binding; the T_max re-auction mechanism.

Tuning before Phase B (§13-T)

- Soft-window width vs duty-cycle; W_a proposal-phase width; max tenure duration; K/K'/K_empty/H_cancel/H_toll_max calibration; Phase A → B criteria.

Read the full story

`docs/preconfirmation/README.md` — index

`status-quo.md` — what exists today, premise validation

`redesign-proposal.md` — the v15 design (this deck's source of truth)

`review-loop/` — rounds 9–14, all dispositions

`simulation/` — model checker + results

`reference/` — the two prior Notion design docs, converted

Discuss

PR **taikoxyz/taiko-mono #22039** — review rounds, defenses, and dispositions are on the thread, in Appendix B, and in `review-loop/`.